Mispriced Tails, Cheap Fees, Uncertain Prices: The Favorite–Longshot Bias in UFC Betting Markets and Its Exploitability Net of Prediction-Market Costs

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Abstract

We test whether mixed martial arts betting markets price fights efficiently, using 6,904 UFC bouts from March 2010 to March 2026 with closing moneyline odds. Through the middle of the probability range the market is strikingly well calibrated, and models built on public pre-fight statistics cannot beat the closing line out of sample — a semi-strong efficiency result. The tails, however, are bent in the direction the favorite–longshot bias literature predicts: heavy underdogs win far less often than their prices imply (bettors backing +300-or-longer underdogs lost 29 cents per dollar staked, nearly nine times the bookmaker margin) and heavy favorites win more. A Mincer–Zarnowitz calibration slope on de-vigged probabilities of 1.114 (fight-clustered SE 0.035) rejects efficiency ($p = 0.001$), survives Shin's insider-trading correction (1.068, $p = 0.046$), and strengthens over time. Exploitability is a stricter claim: a favorites rule selected only on 2010–2017 data earned +1.9% per bet out of sample on 975 bets in 2018–2026 — beating the −4.0% efficient-market benchmark — but not statistically distinguishable from zero ($p = 0.11$). We then price execution on a regulated prediction market using Kalshi's official fee schedule, where the taker fee at heavy-favorite prices is only 0.7–1.8% of stake (1.5% at the sample's mean price). If the exchange mid tracks the de-vigged sportsbook consensus, the same out-of-sample bets return +4.5% per bet ($p < 0.001$), and remain profitable across a one-cent half-spread (+3.3%, $p = 0.01$); the edge dies only if exchange prices embed the bookmaker's margin (+0.6%, $p = 0.35$). A field check on four 2026 cards (49 fights) locates real exchange prices between those poles, and finds the gap depends on *when* one trades: midweek, heavy-favorite asks sit +2.2 cents above the de-vigged consensus, but reconstructed closing quotes show the premium compressing to +0.9 cents — between the frictionless-mid and one-cent-spread scenarios, where the historical edge survives fees. Explicit trading costs are therefore not the binding constraint on exploiting the anomaly — price formation and, above all, depth are.

*rawi.baransy@gmail.com. All results in this paper regenerate from a single command (`python run_analysis.py`) in the accompanying repository; see the Reproducibility statement. This paper is a statistical study of historical prices and is not betting advice.

1 Introduction

When a betting market prices one fighter at −800 and the other at +550, do those prices still reflect true win probabilities? For most of the probability range, betting markets are among the best-calibrated forecasting institutions ever measured. But a persistent exception — the *favorite–longshot bias* (FLB) — has been documented for over seventy years, beginning with horse racing (Griffith, 1949; Ali, 1977; Thaler and Ziemba, 1988): longshots win less often than their prices imply, favorites more often, so the expected return to backing longshots is far worse than the bookmaker's margin alone can explain. Competing explanations divide into preference-based accounts (risk-love or probability weighting by bettors) and belief-based accounts (systematic misperception of small probabilities); Snowberg and Wolfers (2010) show field data favor the misperception account. A separate supply-side mechanism is Shin (1993): bookmakers shade longshot prices to defend against insiders, generating FLB-like patterns even with unbiased bettors — which makes Shin's model the natural conservative benchmark when measuring the bias. For MMA specifically, the evidence is unsettled: in a recent study built on odds aggregated from tapology.com, Miller and Nichols (2026) find no favorite–longshot bias, alongside inefficiencies tied to age and travel. The closing-line sample here reaches the opposite calibration verdict, and Section 4.1 returns to the likely sources of the discrepancy.

Mixed martial arts is a useful laboratory for three reasons. First, every bout is a two-outcome contest with a single dominant bet type (the moneyline), so implied probabilities are clean and the vig is measurable fight by fight. Second, the UFC posts an unusually wide range of prices — pick'ems through −1500 favorites — so the tails are well populated. Third, MMA is young and retail-heavy relative to the horse-racing markets of the classical literature, so it speaks to whether FLB persists in a modern, high-attention sports market.

This paper makes three contributions. (i) It documents the FLB in UFC moneyline markets with formal calibration tests and economic (returns-based) tests, under multiple vig-removal models and with inference that respects the mirrored two-sides-per-fight structure of the data. (ii) It separates *bias* from *exploitability* with a walk-forward protocol that selects a betting rule on an early era and evaluates it blind on a later one, avoiding the data-snooping critique that applies to full-sample strategy returns. (iii) It asks — to our knowledge for the first time in this literature — whether the measured edge survives *prediction-market* execution costs, using the exact fee schedule of Kalshi, the largest CFTC-regulated event-contract exchange, explicit assumptions about exchange price formation, and a first pilot of exchange quotes — live midweek snapshots and reconstructed closing quotes — collected against the multi-book consensus (Wolfers and Zitzewitz, 2004; Kalshi, 2026).

The answer to the exploitability question is sharp but conditional. At average sportsbook closing prices the edge on heavy favorites beats the vig but does not clear zero with statistical confidence. On an exchange, the relevant friction is not a margin inside the price but an explicit fee that is *smallest* exactly where the edge lives: Kalshi's taker fee of $0.07\,P(1-P)$ per contract is 0.7–1.8% of stake at heavy-favorite prices (1.5% at the sample's mean price), versus roughly 4% lost to the vig at a book. Under the standard assumption that a liquid exchange mid tracks the de-vigged book consensus, the out-of-sample favorites rule returns +4.5% per bet ($p < 0.001$) net of fees, and +3.3% ($p = 0.01$) after also crossing a one-cent half-spread. The result reverses only when the assumed exchange price embeds the book's full margin. Fees, in short, are a rounding error on this anomaly; the open question is what prices the exchange actually offers — and the pilot snapshots in Section 4.7 give a first, preliminary reading.

2 Data

The sample is the Ultimate UFC Dataset (mdabbert, 2026), which pairs every UFC bout with closing American moneyline odds and pre-fight career statistics for both corners. After requiring two-sided odds and a plausible book (overround between 1.00 and 1.25, which removes obvious data-entry errors), the sample is 6,904 fights from 2010-03-21 to 2026-03-28. Draws and no-contests (8 bouts, $< 0.15\%$) are excluded from calibration and returns analysis; 6,896 decisive fights remain. The mean overround is 1.0355, implying an efficient-market flat-bet return of $1/r - 1 \approx -3.4\%$: a bettor facing fair de-vigged probabilities loses the vig and nothing more. Clean public odds histories do not exist for other promotions (Bellator, PFL, ONE), so the empirical sample is UFC; the pipeline is promotion-agnostic.

The unit of observation throughout is one *side* of a fight — one price a bettor could take — giving 13,792 sides. The two sides of a fight are mechanically mirrored (if one wins the other loses), so no test in this paper treats them as independent: regressions cluster standard errors by fight, bootstrap confidence intervals resample whole fights, and calibration buckets de-duplicate fights whose two sides land in the same bucket.

A second, much smaller dataset supports the field check of Section 4.7: paired Kalshi-versus-books quotes collected card by card beginning August 2026, at two horizons flagged in the file. *Midweek* rows are live snapshots of Kalshi's UFC order books and the contemporaneous multi-book sportsbook lines, taken several days before each card (two cards, 24 fights; one further bout was dropped because only a single book quoted it and the exchange spread was anomalous). *Close* rows reconstruct the same comparison at fight time for two completed cards (25 fights): the exchange side is the last pre-fight hourly candle from Kalshi's public candlestick history (one to two hours before the market closed), the book side is the aggregator's closing range for each fighter, and — because a binary exchange's two sides share one order book — the underdog quote is the exact complement of the favorite's, which avoids pairing prices read minutes apart. The combined file (49 fights on four cards) is archived in the repository as `data/live_quotes.csv`.

3 Empirical framework

3.1 From prices to probabilities

American odds a convert to decimal odds $d = 1 + a/100$ ($a > 0$) or $d = 1 + 100/|a|$ ($a < 0$), and to raw implied probabilities $p = 1/d$. The two raw probabilities of a fight sum to the booksum $r = p_A + p_B > 1$; $r - 1$ is the bookmaker margin. Because raw probabilities embed the margin, all calibration claims are stated for de-vigged probabilities under two models: (i) *proportional* normalization, $\pi_i = p_i/r$, the standard baseline under which the flat-bet efficient-market return is the same $1/r - 1$ at every price — the property that makes return-by-odds-band comparisons interpretable; and (ii) *Shin* (1993) probabilities, which solve for the insider fraction z per fight and shade longshots relative to proportional scaling. Since Shin's model already attributes part of the raw FLB pattern to bookmaker protection, any bias surviving Shin de-vigging is a conservative finding (Štrumbelj, 2014).

3.2 Hypotheses

The tests are organized as five pre-stated hypotheses.

H1 (Calibration). De-vigged prices equal true win probabilities at every price level. Test: Mincer–Zarnowitz on the logit scale (Mincer and Zarnowitz, 1969), fitting $\Pr(\text{win}) = \sigma(\alpha + \beta \operatorname{logit}(\pi))$ by GLM with fight-clustered standard errors; $H_0 : \alpha = 0, \beta = 1$. $\beta > 1$ means outcomes are more extreme than prices — favorites underpriced, longshots overpriced (the FLB direction). As a nonparametric complement, exact binomial tests per probability decile compare realized win rates with mean implied probabilities, with Benjamini–Hochberg control of the false discovery rate across buckets (Benjamini and Hochberg, 1995).

H2 (Economic calibration). The expected flat-bet return is $1/r - 1$ at every price. Tests: mean returns by American-odds band against the vig-only benchmark with fight-cluster bootstrap intervals (4,000 resamples), and an OLS of per-bet return on implied probability with fight-clustered standard errors, whose slope is zero under efficiency.

H3 (Semi-strong efficiency). Closing prices impound public pre-fight information. Test: models built from 17 public career differentials (record, physical attributes, streaks, striking and grappling averages), fit on fights through 2021 and evaluated on 2022–2026, must not beat the de-vigged market probability on out-of-sample log loss (paired bootstrap over test fights). Fights are oriented favorite-vs-underdog so the arbitrary red/blue corner assignment cannot leak, and features are computed directly from per-corner columns.

H4 (Exploitability at the book). No rule committed to in advance earns ROI > 0 at closing prices. Test: a walk-forward protocol. Eight favorite-cutoff rules (bet every favorite at -150 or shorter, …, -800 or shorter) are evaluated on 2010–2017 only, one is selected by the *most pessimistic* criterion (highest bootstrap lower confidence bound, penalizing small lucky cells), and that single rule is staked blind on 2018–2026. Only the out-of-sample numbers count as evidence.

H5 (Exploitability net of prediction-market costs). The out-of-sample edge from H4 does not survive realistic exchange execution. Test: reprice the same bets as event contracts under Kalshi's official fee schedule (Kalshi, 2026) — taker fee $0.07\,C\,P(1-P)$, maker fee $0.0175\,C\,P(1-P)$ per contract, charged on execution only, with free settlement — under explicit price-formation scenarios described in Section 4.6, and score those scenarios against the live-quote snapshots of Section 4.7.

3.3 Inference conventions

All confidence intervals for returns are percentile intervals from cluster bootstraps that resample whole fights (B = 4,000, fixed seeds); p-values for ROI are one-sided bootstrap probabilities against $H_0 : \text{ROI} \leq 0$ using the recentered bootstrap distribution. Calibration regressions report cluster-robust (fight-level) standard errors. Because the de-vigged probabilities of a fight's two sides are complementary by construction, the M–Z intercept is mechanically zero for de-vigged columns and the slope test is the informative one-degree-of-freedom statistic. No result in this paper depends on treating the 13,792 sides as independent observations.

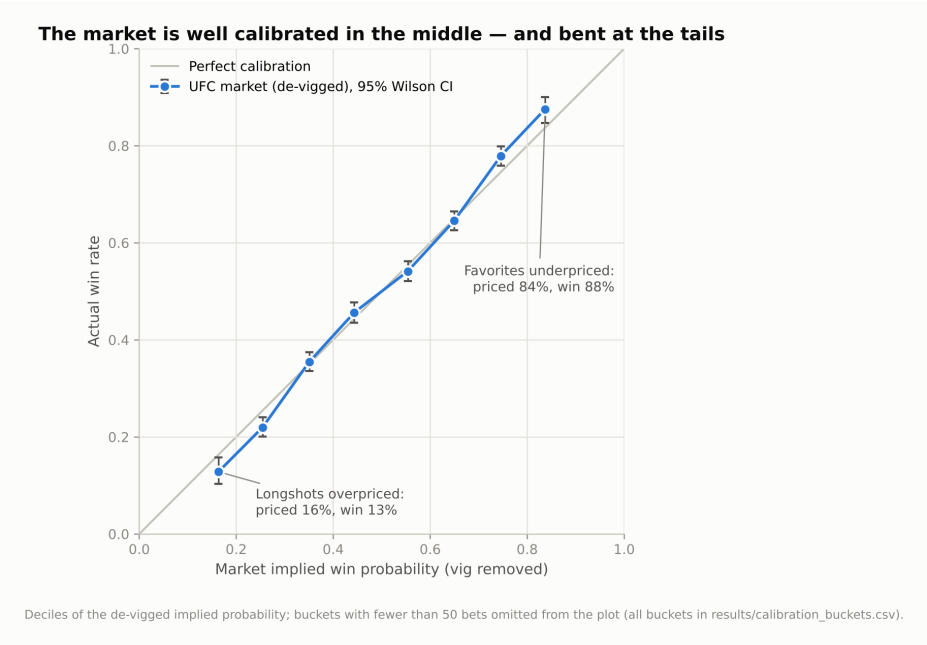


Figure 1: Market calibration on de-vigged (proportional) implied probabilities, probability deciles, 95% Wilson intervals. The market is nearly perfectly calibrated through the middle of the range and bent at both tails in the favorite–longshot direction.

4 Results

4.1 Calibration: efficient middle, bent tails (H1)

Figure 1 shows the calibration curve on de-vigged probabilities in probability deciles with 95% Wilson intervals. Through the middle of the range the market is nearly perfectly calibrated: fights priced 30–70% happen at almost exactly their implied rates (all deviations ≤ 1.3 percentage points, none significant after FDR control). The tails bend in the FLB direction. Sides priced 10–20% (mean 16.3%) won 12.8% of 601 fights; sides priced 20–30% (mean 25.4%) won 22.0% of 1,616 (BH-adjusted $p = 0.009$). On the favorite side, prices of 70–80% (mean 74.6%) won 77.9% of 1,622 (BH $p = 0.009$), and prices of 80–90% (mean 83.7%) won 87.6% of 595 (BH $p = 0.035$). An equal-count robustness binning (27 quantile buckets of roughly 500 bets each) bends identically at the tails.

Table 1 reports the formal test. On proportionally de-vigged probabilities the M–Z slope is 1.114 (SE 0.035), rejecting $H_0 : \beta = 1$ at $p = 0.001$. The estimate is above one for raw probabilities as well (1.063, $p = 0.061$; the margin itself masks part of the tail distortion), and — the conservative headline — remains above one under Shin de-vigging (1.068, $p = 0.046$). H1 is rejected: prices are too timid at the extremes even after allowing for insider-defensive price shading.

This verdict differs from Miller and Nichols (2026), who find no favorite–longshot bias in MMA using odds sourced via tapology.com. Three design differences plausibly account for the gap. The odds provenance differs: this sample is a single aggregated *closing* line per fight, whereas listed odds collected earlier in the price-discovery process, or from a different aggregation, embed different margins and different information sets. The tests differ: the logit-

Table 1: Mincer–Zarnowitz calibration tests, 13,792 sides of 6,896 fights.

Probability measure	Slope $\hat{\beta}$	SE (fight-clustered)	p for $H_0\colon \beta = 1$
Raw implied (p)	1.063	0.034	0.061
De-vigged, proportional (π)	1.114	0.035	0.001
De-vigged, Shin (1993)	1.068	0.034	0.046

Logit-scale GLM $\Pr(\text{win}) = \sigma(\alpha + \beta \operatorname{logit}(\hat{p}))$. $\beta > 1$: outcomes more extreme than prices (favorites underpriced, longshots overpriced). For de-vigged measures the intercept is zero by construction and the slope test carries the content.

scale M–Z regression with fight-clustered standard errors concentrates power on exactly the tail curvature at issue, and the economic profile of Table 2 — a −27 to −37 cent per-dollar flat-bet loss on heavy underdogs against a four-cent margin — is difficult to generate without tail mispricing. And the samples differ in span and composition. Adjudicating between the two results on a common dataset is a natural follow-up; at minimum, the disagreement shows the MMA evidence is sensitive to odds provenance, which argues for the closing line as the cleanest benchmark.

4.2 The economic version: returns by price (H2)

Calibration gaps of a few probability points at the tails translate into enormous return asymmetries because longshot payouts lever the error. Table 2 and Figure 2 report mean flat-bet returns by odds band against the −3.4% vig-only benchmark. Underdogs at +300 to +499 lost 26.8% of stake per bet (95% CI $[-37.4, -15.3]$, $n = 937$); underdogs at +500 or longer lost 37.0% $[-59.7, -12.4]$ ($n = 282$). No bookmaker margin explains losing 27–37 cents per dollar when the margin is under four cents: this is probability mispricing. On the other side, every favorite band from −300 on beats the benchmark, and −500 to −799 favorites returned +2.8% $[-0.8, +6.5]$ against a −3.5% null (bootstrap $p = 0.002$ against the null). The OLS of per-bet return on implied probability has slope +0.258 (fight-clustered $t = 3.76$, $p < 0.001$): every additional point of implied probability raises expected return — shorter prices are systematically better bets, the economic signature of FLB. Cumulatively (Figure 3), flat-betting every −300-or-shorter favorite over sixteen years ended +26 units on 1,804 bets *through the vig*, while flat-betting every +300-or-longer underdog lost 355 units on 1,219 bets.

4.3 The bias is not an early-market relic

Re-estimating the M–Z slope by era gives 1.03 (SE 0.06) for 2010–2015, 1.11 (0.06) for 2016–2020, and 1.21 (0.06) for 2021–2026 — the latest era rejects efficiency on its own ($p < 0.001$), and the point estimates *rise* across eras (Figure 4). Whatever generates the mispricing has not been competed away; it has grown with the sport's mainstream (and recreational-bettor) audience, consistent with demand-side accounts of FLB (Snowberg and Wolfers, 2010) and difficult to reconcile with early-market naivety.

Table 2: Flat-bet returns by American-odds band, 2010–2026.

Band	n	Implied (de-vigged)	Won	Return	95% CI	p vs. vig null
−800 or shorter	175	0.877	0.909	−0.3%	$[-5.2, +4.2]$	0.14
−799 to −500	408	0.826	0.880	+2.8%	$[-0.8, +6.5]$	0.002
−499 to −300	1,221	0.759	0.795	+1.2%	$[-1.6, +4.2]$	0.002
−299 to −200	1,806	0.679	0.686	−2.5%	$[-5.5, +0.6]$	0.53
−199 to −140	2,050	0.600	0.585	−5.6%	$[-8.9, -2.3]$	0.18
−139 to +99	1,576	0.522	0.511	−5.9%	$[-9.3, -2.6]$	0.24
+100 to +139	1,746	0.443	0.458	−0.4%	$[-5.5, +4.6]$	0.22
+140 to +199	2,054	0.366	0.370	−2.5%	$[-8.1, +3.2]$	0.76
+200 to +299	1,537	0.286	0.281	−5.4%	$[-13.0, +2.4]$	0.61
+300 to +499	937	0.208	0.156	−26.8%	$[-37.4, -15.3]$	< 0.001
+500 or longer	282	0.135	0.089	−37.0%	$[-59.7, -12.4]$	0.001

One bet per listed price (both sides of every fight). CIs: fight-cluster bootstrap, $B = 4{,}000$. Vig-only null: $1/r - 1 \approx -3.4\%$ in every band under proportional margin; p is the two-sided bootstrap p-value against that null.

Table 3: Out-of-sample benchmark vs. the closing line, 2022–2026 ($n = 1{,}882$ fights).

Model	Log loss	Brier	AUC	Δ log loss vs. market [95% CI]
Market (de-vigged odds)	0.588	0.201	0.642	—
Stats only (logistic)	0.606	0.208	0.583	$+0.018$ $[+0.009, +0.027]$
Stats + market (logistic)	0.585	0.201	0.645	-0.003 $[-0.008, +0.002]$
Stats + market (grad. boost)	0.598	0.206	0.634	$+0.010$ $[-0.000, +0.020]$

Chronological split (train $\leq$ 2021). Negative Δ = model beats the market. CIs: paired bootstrap over test fights. Features are favorite-minus-underdog career differentials computed from per-corner columns; rows oriented favorite-vs-underdog so corner assignment cannot leak.

4.4 Semi-strong efficiency: public information cannot beat the line (H3)

If the tails are mispriced, does the market also ignore public information? No. Table 3 reports the out-of-sample benchmark on 1,882 fights from 2022–2026. A logistic model on 17 public career differentials alone is *much* worse than the market (log-loss gap $+0.0178$, 95% CI $[+0.009, +0.027]$). Adding the market probability as a feature recovers the market almost exactly and improves on it by a statistically indistinguishable -0.0027 $[-0.0075, +0.0021]$; gradient boosting overfits and loses. A betting simulation that backs whichever side the combined model prices above the raw market price never achieves a significant ROI at any edge threshold (point estimates -0.8% to $+1.8\%$, minimum one-sided $p = 0.40$). H3 survives: the anomaly is a systematic distortion of tail prices, not exploitable neglect of public statistics — mispricing and informational efficiency coexist.

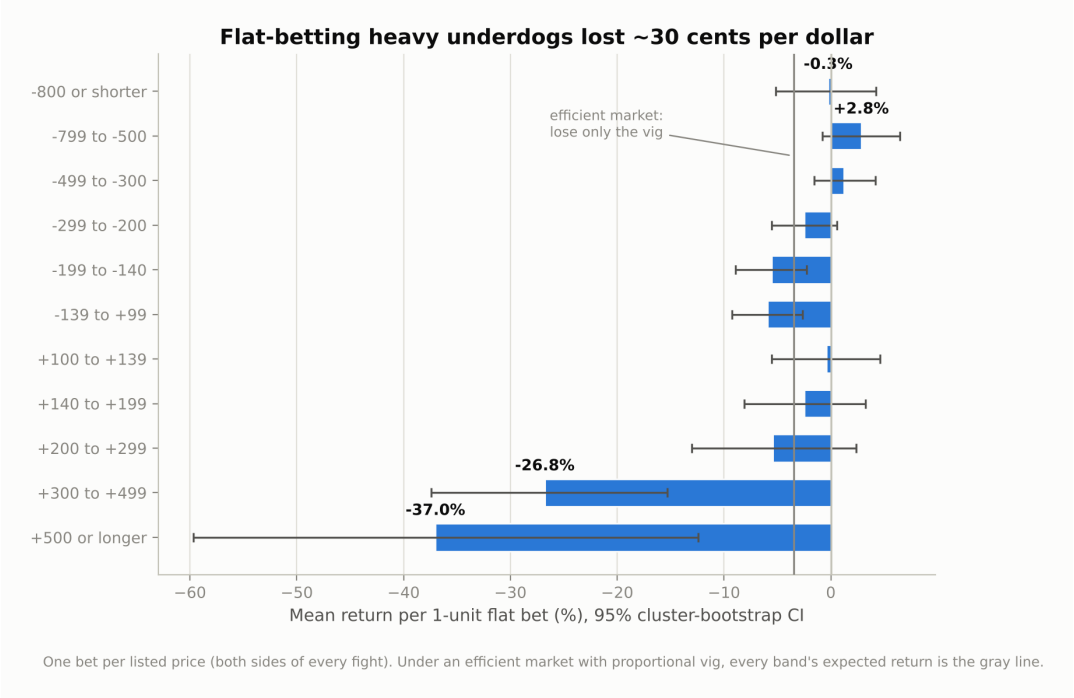


Figure 2: Mean return per 1-unit flat bet by odds band with 95% cluster-bootstrap intervals. Under efficiency every band's expected return equals the vig-only benchmark (gray line).

4.5 Exploitability at the sportsbook: real bias, marginal profit (H4)

Bias and exploitability are different claims, and the second demands a stricter test: scanning sixteen years for the best-looking rule and reporting its return is data snooping. In the walk-forward protocol, the rule chosen on 2010–2017 alone (by highest bootstrap lower bound — the pessimistic criterion) is "flat-bet every favorite at −300 or shorter," with training ROI +1.0% on 829 bets. Staked blind on 2018–2026 it returned +**1.9**% per bet on 975 bets (+18.1 units; 95% CI $[-1.2, +4.7]$), versus the −4.0% an efficient market implies for those prices (Figure 5). The bias therefore did not evaporate after its "discovery" in training data — the rule beat the efficient-market benchmark by roughly six points of ROI — but its profit is not statistically distinguishable from zero (one-sided $p = 0.11$). At average closing prices, H4 is not rejected: the tradeable content of the anomaly at a sportsbook is mostly *avoidance* (never flat-bet longshots), not a reliable money pump.

4.6 Execution on a prediction market: fees are not the constraint (H5)

The preceding test prices execution at a sportsbook, where the friction — the overround — is baked into the price. Event-contract exchanges change the cost structure entirely. On Kalshi, a YES contract on the favorite trades at a price $P \in (0, 1)$ dollars and pays \$1 at settlement; there is no vig in the price, and the explicit costs are a trading fee and the bid–ask spread. Kalshi's fee schedule (effective July 7, 2026) charges, per contract and only on execution (settlement, membership, and ACH transfers are free):

$$\text{taker fee} = 0.07\,P(1-P), \qquad \text{maker fee} = 0.0175\,P(1-P).$$

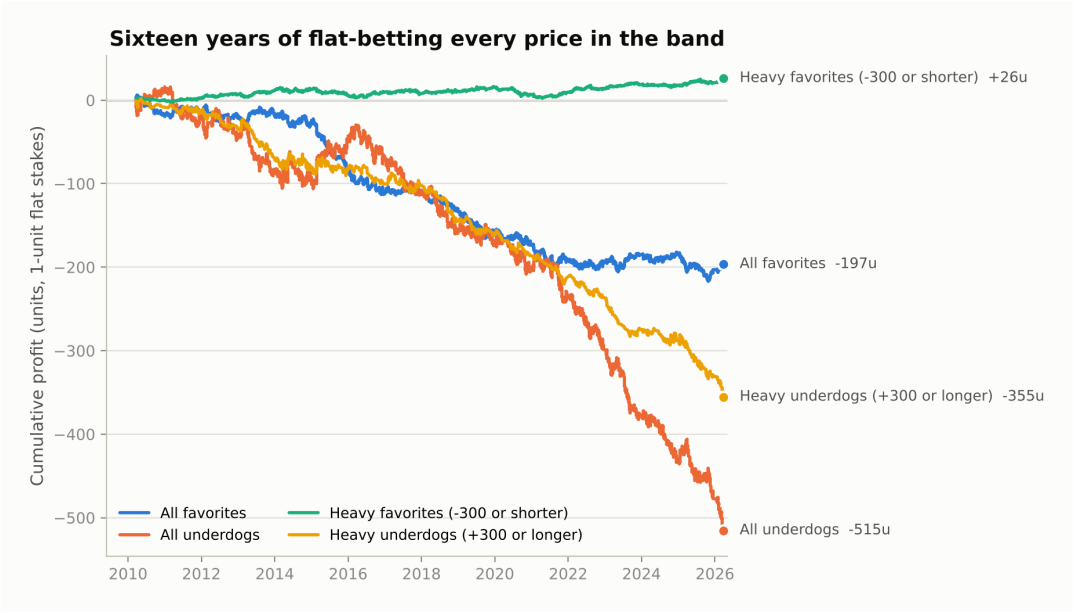


Figure 3: Cumulative profit of 1-unit flat stakes, 2010–2026. Heavy favorites (−300 or shorter) end +26 units through the vig; heavy underdogs (+300 or longer) lose 355 units.

The fee is a parabola peaking at $P = \$0.50$ (1.75 cents). The edge documented in this paper lives at $P \approx 0.75$–0.90, where the taker fee is 0.6–1.3 cents per contract — as a share of the price paid, essentially $0.07(1-P)$, i.e. 0.7–1.8%, *falling* in the favorite's price. The venue that appears expensive for coin-flip contracts is nearly free precisely where the FLB edge sits, while the sportsbook's effective margin (∼4% of stake) does not shrink for favorites.

Kalshi listed its first UFC markets in 2025, so no historical exchange prices exist for this sample, and H5 cannot be tested against realized exchange quotes over 2010–2026. We therefore model execution under explicit price scenarios spanning optimistic to worst-case, and then score those scenarios against live quotes in Section 4.7. Staking \$1 buys $1/(P + \text{fee})$ contracts; the per-dollar return is won$/(P + \text{fee}) - 1$. Under (price = raw closing implied probability, fee = 0) this accounting reproduces the sportsbook flat-bet return exactly — a unit-tested identity, so the scenarios generalize the baseline rather than changing the measurement.

Table 4 and Figure 6 give the result. If the exchange mid tracks the de-vigged consensus of the books — the standard assumption for liquid, arbitrage-linked sports markets — the identical bets that were marginal at the book clear zero decisively: +4.5% per bet as a taker at the mid ($p < 0.001$), because the ∼4 points the book kept as vig accrue to the bettor while the exchange charges only 1.5%. The edge survives crossing a one-cent half-spread (+3.3%, $p = 0.012$), thins at two cents (+2.1%, $p = 0.076$), and disappears in the worst case where the exchange price embeds the book's full margin (+0.6%, $p = 0.35$). The full-sample version of the rule (1,804 bets, 2010–2026) shows the same ordering (mid taker +3.5%, $p = 0.002$; one-cent spread +2.2%, $p = 0.022$; two cents +1.0%, $p = 0.17$), as does the recent era alone (2021–2026, 650 bets: mid taker +5.8%, $p < 0.001$; even the *sportsbook* baseline is positive there at +3.0%, $p = 0.04$, though selecting the strongest era after observing it is descriptive, not confirmatory).

A breakeven decomposition makes the economics transparent. Out of sample the favorites

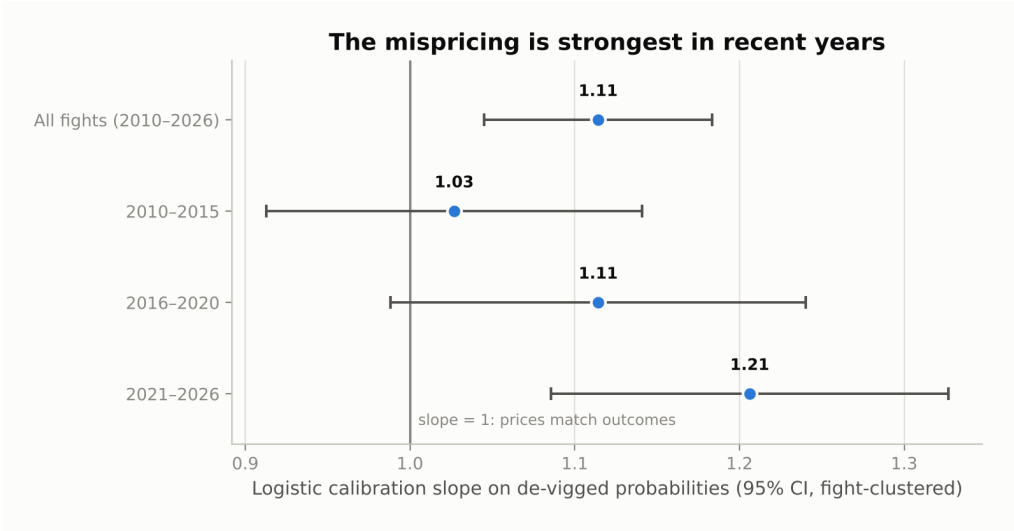


Figure 4: Calibration slope by era with 95% intervals (fight-clustered). The mispricing is strongest in the most recent era.

Table 4: The walk-forward out-of-sample bets (favorites ≤ -300, 2018–2026, $n = 975$) under execution scenarios.

Scenario	Price basis	Fee (% of stake)	ROI	95% CI	p
Sportsbook closing line (baseline)	raw close	—	+1.9%	$[-1.2, +4.7]$	0.10
Kalshi taker at the mid	de-vigged close	1.5%	+4.5%	$[+1.4, +7.4]$	< 0.001
Kalshi taker, mid + 1¢	de-vigged + .01	1.4%	+3.3%	$[+0.2, +6.1]$	0.012
Kalshi taker, mid + 2¢	de-vigged + .02	1.4%	+2.1%	$[-1.0, +4.9]$	0.076
Kalshi maker at the mid	de-vigged close	0.4%	+5.7%	$[+2.5, +8.6]$	< 0.001
Kalshi taker at book price (worst)	raw close	1.3%	+0.6%	$[-2.4, +3.3]$	0.35

Fees per Kalshi's official schedule; exact formula without per-order rounding (rounding adds at most one cent per order). CIs: fight-cluster bootstrap; p: one-sided against ROI ≤ 0. Mean prices: raw close \$0.816, de-vigged mid \$0.783. The baseline row recomputes the Section 4.5 bets inside the cost model's bootstrap; its p of 0.10 vs. 0.11 there is seed variation at $B = 4{,}000$. The maker row ignores fill risk and adverse selection and is an upper bound.

won 83.2% of fights against a mean de-vigged mid of 78.3%: an edge of 4.9 probability points. Total friction (fee plus effective half-spread) of up to 6.2% of the mid price would be required to erase it. The Kalshi taker fee consumes 1.5%, leaving roughly 3.7 cents per contract of headroom for the spread — several times the tick width quoted on liquid sports markets. This is why the verdict is insensitive to the fee and hinges entirely on price formation: *whether* Kalshi's UFC mids actually sit at the de-vigged book consensus, at book-equivalent prices, or — if exchange participants correct the bias — at the true probability, at which point no venue rescues the strategy. Evidence that prediction markets exhibit their own favorite–longshot bias (Page and Clemen, 2013) makes the consensus-tracking scenario plausible a priori; the next section brings the first direct observations to bear.

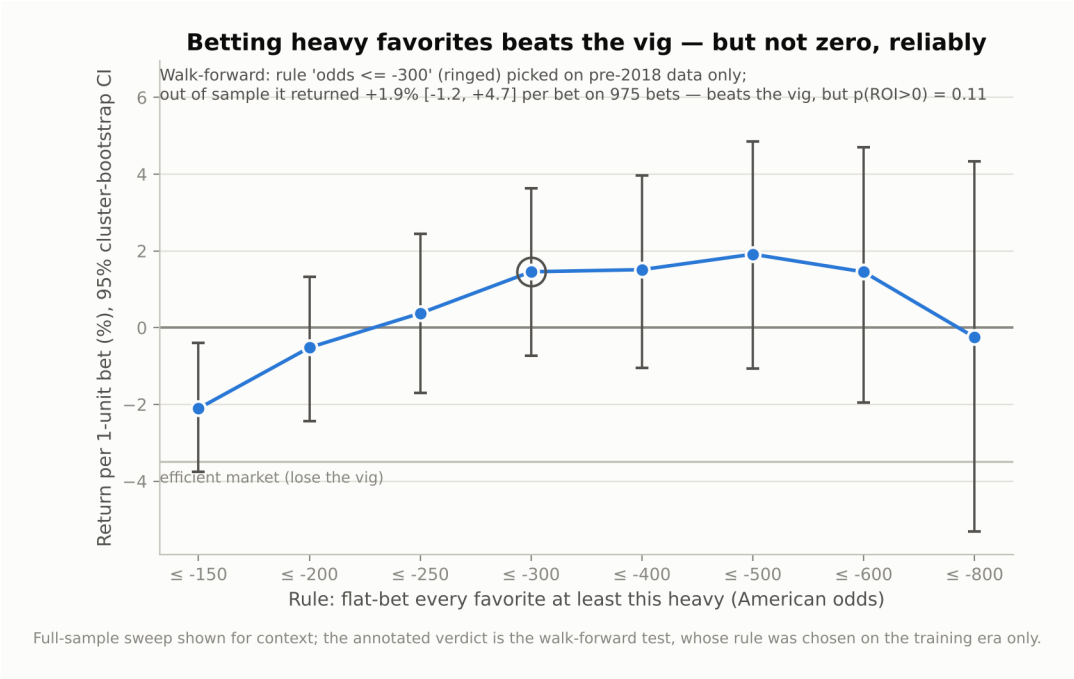


Figure 5: Favorite-cutoff sweep (full sample, for context) and the walk-forward verdict. The ringed rule was selected on 2010–2017 data only and staked blind on 2018–2026.

4.7 Early field evidence: where live exchange prices actually sit

Which scenario row describes reality is an empirical question, and since Kalshi now lists every UFC bout, it can be answered by direct observation. I collect the comparison at two horizons. For upcoming cards, live midweek snapshots pair Kalshi's best bid and ask from the public order book with the contemporaneous moneyline range across seven sportsbooks from a multi-book aggregator (two cards: August 29 and November 1, 2026; 24 fights). For completed cards, the same comparison is reconstructed *at fight time*: Kalshi's public candlestick history preserves the order book's hourly bid and ask, so the exchange quote is the last pre-fight hourly candle (one to two hours before the recorded market close), paired with the aggregator's closing range for each fighter (two cards: UFC 331 on September 19 and the Edmonton card of October 17, 2026; 25 fights). In both designs the consensus is approximated as the probability-space midpoint of each side's best and worst book price, de-vigged proportionally; the statistic of interest is the favorite's ask minus the de-vigged consensus — what a taker actually pays above the "mid tracks consensus" benchmark.

Three observations stand out (Figure 7; per-fight table in `results/field_quote_gaps.csv`). First, the exchange microstructure is as cheap as the fee analysis assumed, at both horizons: quoted spreads average 1.0–1.2 cents, and the two sides' asks sum to roughly 1.00–1.01 — there is essentially no vig in the prices. Second, favorite asks sit *above* the de-vigged consensus, but by how much depends on when one looks. Midweek, the heavy favorites the strategy would trade cost +2.2 cents over the consensus (+1.6 at the mid; 8 heavy favorites). At close — the horizon on which the historical edge is actually defined — the premium compresses to +0.9 cents at the ask and +0.4 at the mid (10 heavy favorites), and the compression holds across all fights as well (+1.7 midweek vs. +0.9 at close). Third, the pattern is consistent within horizon

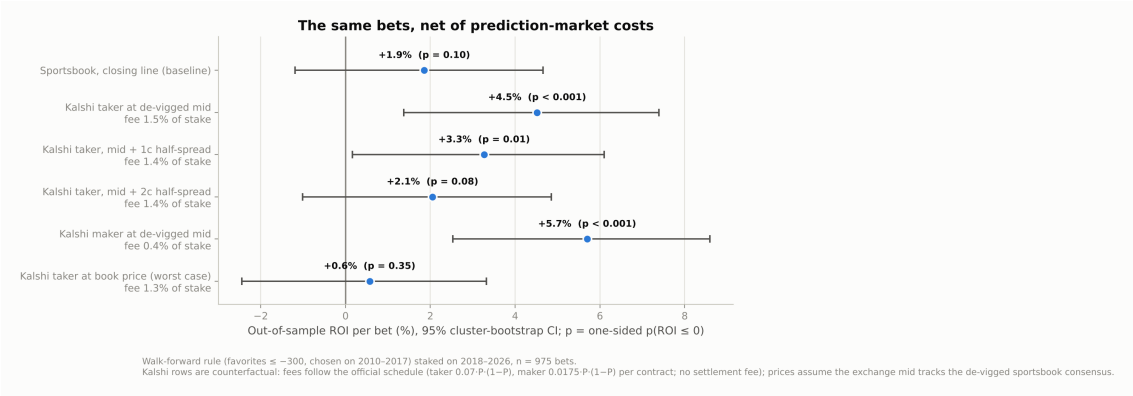


Figure 6: Out-of-sample ROI of the walk-forward rule under execution scenarios, with 95% cluster-bootstrap intervals and one-sided p-values. Kalshi rows are counterfactual: fees follow the official schedule; prices are assumptions about exchange price formation.

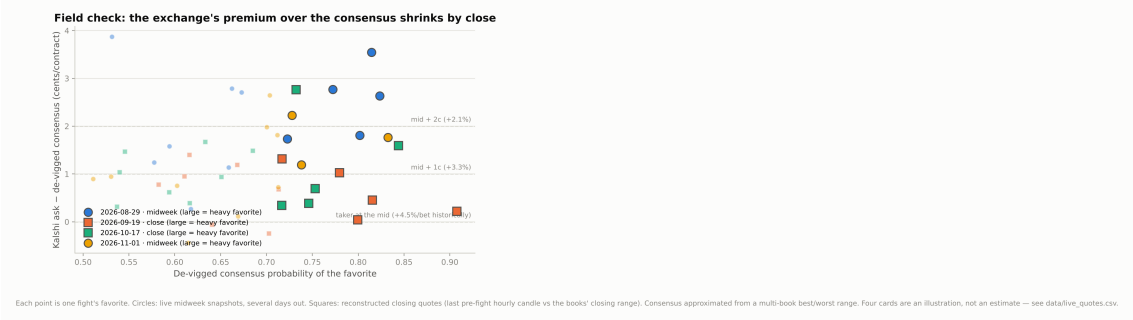


Figure 7: Pilot field check: each point is one fight's favorite; the vertical position is the Kalshi ask minus the de-vigged multi-book consensus, with the cost model's scenario rows as dashed references. Circles are live midweek snapshots; squares are reconstructed closing quotes (last pre-fight hourly candle vs. the books' closing range). Large markers are consensus favorites of −300 or shorter — the subset the walk-forward strategy trades.

across cards rather than an artifact of one market. Mapped onto Table 4, closing-time taker execution lands between the frictionless-mid row (+4.5% per bet historically, $p = 0.001$) and the one-cent row (+3.3%, $p = 0.012$) — a materially better regime than the midweek two-cent gap, which corresponds to the +2.1% row ($p = 0.076$). Read causally, early exchange prices leave favorites rich and converge toward the book consensus as money arrives before the fight; but the two horizons were measured on *different* cards, so this is a cross-card contrast, not a within-card price path. Resting bids one tick below the ask would execute at or below the consensus itself, between the maker and taker rows, with fill risk this data cannot measure.

These are pilot numbers, to be treated as an illustration rather than an estimate: two cards per horizon, a consensus proxied from a best/worst range rather than the full quote list, and closing quotes taken from the last hourly candle rather than the final tick (book closing ranges likewise aggregate books that stop updating at slightly different times). The collection design that would settle H5 — the same closing-horizon reconstruction over 20–30 cards, roughly 50 heavy favorites — pins the mean closing gap to within about half a cent and is underway; the repository's `field` module ingests each new card as CSV rows and regenerates this analysis.

5 Verification and robustness audit

Because this study makes an economic claim, the analysis was audited end to end before results were interpreted. The audit covered five classes of error.

Reproducibility. The entire pipeline — data filters, tests, tables, figures — runs from one command with fixed random seeds, and every number in this paper regenerates bit-identically from the frozen dataset snapshot (one library-version caveat for the gradient-boosting row is noted in the Reproducibility statement). The repository ships 33 unit tests covering the statistical machinery.

Statistical dependence. The two sides of a fight are mirror images. Every regression clusters by fight; every bootstrap resamples whole fights; calibration buckets de-duplicate mirrored pairs (alternating which side is kept, since always keeping one corner would inject the corner-assignment effect into the pick'em bucket). The unit tests include power checks on synthetic markets: a calibrated market ($\beta = 1$) produces no rejection and no walk-forward edge; a market simulated with $\beta = 1.35$ produces both.

Vig contamination. All headline claims are stated on de-vigged probabilities; the bias survives the conservative Shin correction, whose insider-defense mechanism would otherwise be the leading supply-side confound. The economic tests use a benchmark ($1/r - 1$) that is price-invariant under proportional margin, so band comparisons are apples to apples.

Data snooping. The exploitability claim rests solely on a rule selected on 2010–2017 by a pessimistic criterion and evaluated once on 2018–2026. The full-sample sweep and the recent-era slice are reported as context and labeled descriptive. Per-bucket calibration p-values carry FDR control.

Look-ahead and leakage in the ML benchmark. Chronological split; features built only from pre-fight career columns; fights oriented favorite-vs-underdog so corner labels cannot leak; the dataset's own pre-computed differential columns (which mix orientations) are not used. The fee-scenario accounting was validated by an identity test: with price set to the raw close and zero fee it reproduces the sportsbook return series exactly. The field module's gap computation is validated the same way: synthetic snapshots constructed backwards from a known gap must return exactly that gap.

6 Limitations

Five limitations bound the conclusions. First, odds are closing lines from a single aggregated source; intraday movement, best-price line shopping across books, and bet limits are outside the data. Second, the exchange analysis of Section 4.6 is counterfactual for the historical sample: Kalshi listed no UFC markets before 2025, so the scenario rows cannot be tested against realized 2010–2026 exchange quotes. The field data of Section 4.7 begin to discipline the price assumption — including at the closing horizon, where the reconstructed quotes sit within a cent of the consensus — but four cards split across two horizons are a pilot, the midweek-to-close compression is a contrast between different cards rather than a within-card price path, and exchange depth at 80-cent-plus prices, position limits, and maker fill rates all remain unmeasured; the maker row in particular is an upper bound that ignores adverse selection. Third, although the walk-forward protocol guards against rule snooping, the broader research design (studying favorites because the FLB literature predicts favorites) is itself informed by prior evidence; the protection is the out-of-sample era, not ignorance of the hypothesis. Fourth,

results are UFC-specific; other promotions' markets may differ, and Miller and Nichols (2026) show that MMA conclusions can differ with odds provenance. Fifth, the profitable cell is concentrated: heavy favorites win 83% of the time, so the strategy's returns are a long series of small gains punctuated by -100% losses, and its five-point probability edge is real money only at stake sizes the observed order books — open interest of a few hundred to a few tens of thousands of dollars per side — may not absorb.

7 Conclusion

UFC moneyline markets from 2010–2026 present a sharp two-sided picture. They are informationally efficient — superbly calibrated through the middle of the probability range, impounding public fighter statistics so completely that no model built on them beats the closing line — and yet systematically mispriced at the tails, in the classic favorite–longshot direction, by margins that are economically enormous on the longshot side (a 29-cent-per-dollar flat-bet loss on heavy underdogs) and statistically decisive throughout (calibration slope 1.114, $p = 0.001$, robust to Shin de-vigging, strongest in the most recent era).

On the question this study set out to answer — *is there an exploitable edge once prediction-market costs are accounted for?* — the answer is: **the costs are not the obstacle**. At sportsbook closing prices the pre-registered favorites rule beat the efficient-market benchmark by six points of ROI out of sample but could not clear zero statistically (+1.9%, $p = 0.11$), because the book's margin consumes most of the mispricing. Kalshi's fee at heavy-favorite prices is roughly a third of that margin at the sample's mean price, and settlement is free; under the standard assumption that exchange mids track the de-vigged book consensus, the same bets return +4.5% per bet ($p < 0.001$) net of fees, surviving a realistic spread. What the fee schedule cannot guarantee is the price. The first direct observations — two live midweek cards and two cards of reconstructed closing quotes — find that where exchange prices sit depends on the horizon: spreads are tight and the vig is gone throughout, but heavy-favorite asks that stand two cents above the de-vigged consensus midweek compress to within about one cent by fight time. Taken at face value, closing-time taker execution lands between the frictionless-mid and one-cent scenarios — the regime where the historical edge survives fees — though four pilot cards cannot yet bear the weight of that conclusion. The binding constraints on exploiting a seventy-year-old anomaly in a modern market are price formation and, above all, depth, not explicit transaction costs. Extending the closing-quote reconstruction across 20–30 cards — and, if the gap stays small, testing maker execution with its fill risk — converts H5 from scenario analysis into measurement, and is the project's next step.

Reproducibility statement

All results derive from a pinned snapshot of the public Ultimate UFC Dataset (6,904 fights, 2010-03-21 to 2026-03-28) and regenerate bit-identically via `python run_analysis.py` (fixed seeds, $B = 4{,}000$ bootstrap resamples throughout; 33 passing unit tests via `pytest`), with one caveat: the gradient-boosting benchmark row varies at the third decimal with the scikit-learn version, because the estimator's internals evolve across releases even at a fixed seed. Committed results were generated with numpy 2.5.3, pandas 3.0.5, scipy 1.18.1, statsmodels 0.15.0, scikit-learn 1.9.1, and matplotlib 3.11.2. The repository contains the full pipeline: odds conversion

and de-vigging (proportional, power, Shin), calibration and returns tests, the walk-forward protocol, the ML benchmark, the Kalshi fee model (`src/mma_efficiency/costs.py`), and the live-quote field analysis (`src/mma_efficiency/field.py` with `data/live_quotes.csv`), together with all tables (`results/`) and figures (`figures/`) used here.

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